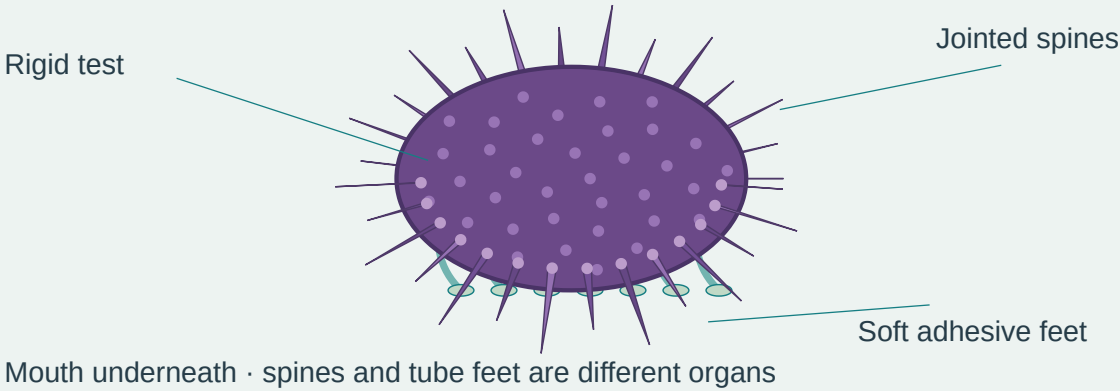


Sea urchins

A slow crawl built from many small contacts.

SHORT-SPINED REGULAR URCHIN



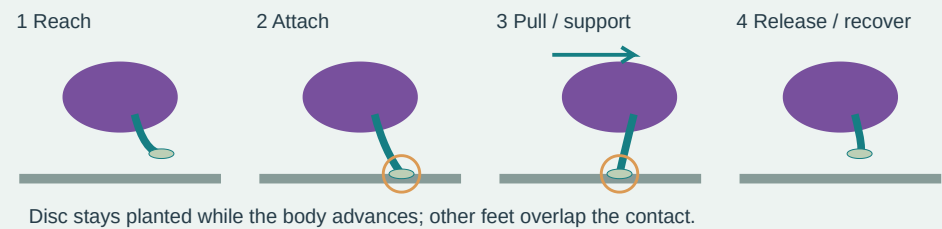
Two kinds of moving parts

The test is the inner shell. Purple urchins move with tube feet; their spines pivot on ball-and-socket joints. The mouth lies underneath. [1]

Strongylocentrotus purpuratus is our proposed visual reference: a short-spined resident of eastern Pacific rocky shores, with a test up to about 7 cm across. [1]

Original schematics, not photographs. Different species supply different evidence below.

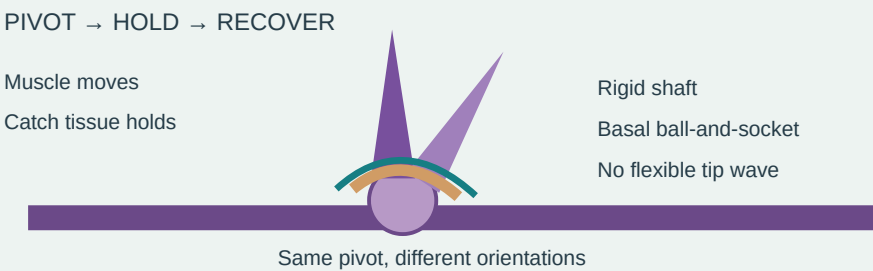
01 · Attach, pull, let go



Tube feet make and release adhesive contacts; spines also assist movement. The discs use secretions rather than simple suction. [2,5]

The sequence above is an animation abstraction. Overlapping contacts should hold their world position as the body passes.

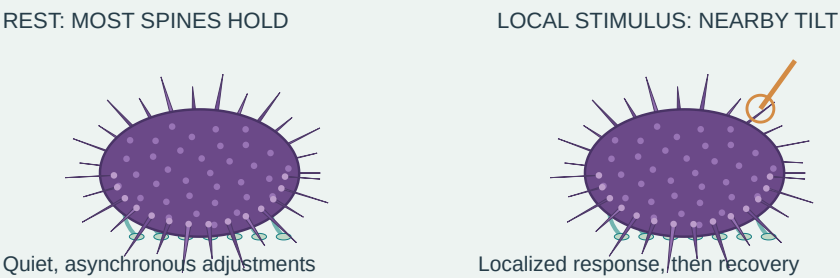
02 · Tilt at the base; hold the pose



Muscle moves a spine. Mutable collagenous catch tissue changes stiffness and helps maintain its orientation. [3]

Animate the basal joint and rigid shaft. Soft tentacle-like bending would give the wrong mechanical impression.

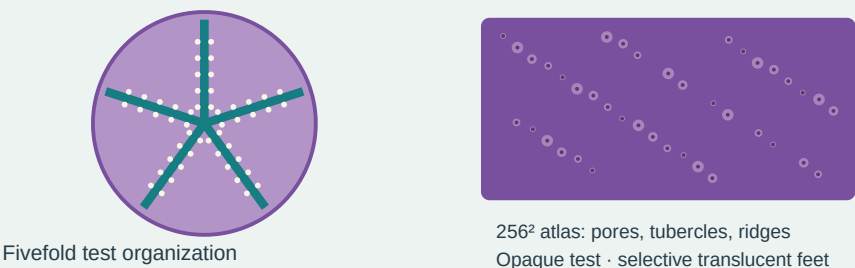
03 · A response can be local



In *Diadema setosum*, touch coordinates nearby spines; a shadow can trigger waving. [4]

Do not transfer that shadow reflex to every species. For our short-spined model, sparse adjustments and held poses are a proposed starting point.

04 · Texture supports the anatomy



The test has fivefold organization. [6] Distinguish shaft bases, fine secondary spines and soft foot discs.

Game proposal: 256² atlas, surface markings and ridges; selective foot translucency. Geometry carries the silhouette.

05 · Small Forth ideas for planted feet

```
\ Keep a planted disc in world space
: contact-local ( anchor body -- offset ) - ;
\ Bounded response; tau must be positive
: lag-step ( x target dt tau -- x' )
  / 1 min 0 max >r over - r> * + ;
```

Move slowly in the requested substrate direction. Advance traction from actual travel; blocked input must not make the animal skate.

Keep tap A for camera and hold A for plant settings. Separate foot traction, exploratory feet and local spine response.

Illustrative fragments. Angles, timing and speed are game choices to tune from footage and measure on hardware.

Separate research: [sea-urchin-research.md](#) · Checked 5 October 2026.

[1] [Purple urchin · Monterey Bay Aquarium](#) · [2] [Tube-foot secretion · Pjeta et al. \(2020\)](#) · [3] [Catch apparatus · Takemae & Motokawa \(2005\)](#) · [4] [Spine coordination · Motokawa & Fuchigami \(2015\)](#) · [5] [Adhesion and movement · Garner et al. \(2024\)](#) · [6] [Fivefold test · ODFW](#)

A3 portrait · 297 × 420 mm · Marine animals; freshwater is not a suitable habitat. Sources name their study species; pose drawings are conceptual, not measured kinematic traces.